

## Worksheet — Motion in a Plane (NIOS Official)

Total marks: 37 · Questions: 10

Instructions: This is the NIOS Senior Secondary Physics (312) official Worksheet-4 for Lesson 4. Attempt all questions. Most are open-ended and inquiry-based — observe your surroundings, describe what you see, and explain in your own words. Show your steps for numerical questions. (Take  $g = 10 \text{ m/s}^2$  unless otherwise stated.)

Q1. Observe your surroundings and give examples of projectile motion from day to day life. <font color='#888888'>[2 marks]</font>

Q2. Take two balls and project one of them horizontally from the top of a building. At the same time drop the other ball downward from the same height. Observe the path of both balls and the time to hit the ground. Is the time the same? Explain with reasons. <font color='#888888'>[3 marks]</font>

Q3. Take a ball and project it horizontally from the top of a building. Observe the projectile motion of the ball and explain with reasons which two quantities are constant throughout the projectile motion (when air resistance is negligible). <font color='#888888'>[3 marks]</font>

Q4. Suppose you want to launch a projectile to land at a certain target (for instance, a cricket ball beyond the boundary). What are the important factors you have to calculate before launching the projectile? Analyse projectile motion to determine all these factors. <font color='#888888'>[5 marks]</font>

Q5. Suppose you are riding a bicycle along a circular path of the park near your house. Measure the radius of the circular path. Count the time to complete one round and calculate the total length of the circular path. <font color='#888888'>[3 marks]</font>

Q6. Take three stones and throw them at different angles such that they reach the same maximum height. Observe the trajectories of all three projectiles and calculate for all three (a) time of flight, (b) horizontal velocity component, (c) downward acceleration component. Are these the same for all three stones? Explain. <font color='#888888'>[5 marks]</font>

Q7. Derive the equation of the path of projectile motion and the equation of the trajectory. <font color='#888888'>[5 marks]</font>

Q8. Observe your surroundings and give examples of uniform circular motion from day to day life. Explain applications of uniform circular motion in day to day life. <font color='#888888'>[3 marks]</font>

Q9. A truck goes around a circular track of radius  $R$  at speed  $v$  so that it makes one circuit every  $T$  seconds. It experiences a centripetal acceleration of magnitude  $a$ . If the truck now goes around a different circular track of radius  $4R$ , so that it takes time  $T/2$  to go around once, what is the magnitude of its centripetal acceleration in the second case? [3 marks]

Q10. Explain in your own words what is meant by centripetal acceleration. Derive an expression for the same. [5 marks]

# Answer Key

Worksheet — Motion in a Plane (NIOS Official)

Q1. Observe your surroundings and give examples of projectile motion from day to day life.

**Answer: Examples of projectile motion: (1) A cricket ball hit for a six — launched at some angle, follows a parabolic path until caught or hits the ground. (2) A football kicked for a goal — same parabolic flight. (3) Water from a garden hose held at an upward angle. (4) A stone thrown to break a coconut from a tree. (5) A long-jumper's centre-of-mass during the leap. (6) Water from a fountain. (7) An arrow shot from a bow. (8) A bullet fired horizontally falls in a parabola (not straight). All are characterised by a constant horizontal velocity component and a constant downward acceleration  $g$  due to gravity (air resistance neglected). The path is a parabola.**

Look for at least three valid examples — anything launched into the air and acted on only by gravity.

SHORT\_ANSWER · EASY · 12-NIOS-PHY-MIP-06

Q2. Take two balls and project one of them horizontally from the top of a building. At the same time drop the other ball downward from the same height. Observe the path of both balls and the time to hit the ground. Is the time the same? Explain with reasons.

**Answer: Observation: BOTH balls hit the ground at the SAME TIME, even though one is moving horizontally and the other is only falling vertically. Reason: in projectile motion the horizontal and vertical components of motion are independent. The horizontal motion does not affect the vertical motion. Both balls have the same vertical initial velocity (zero) and experience the same vertical acceleration  $g$  downward. So they take the same time  $t = \sqrt{2h/g}$  to fall the height  $h$ . The difference: the horizontally-launched ball travels some horizontal distance  $R = u \cdot t$  during its fall and lands away from the building. The dropped ball lands directly below. But the TIME of fall is the same. This is a classic demonstration that the two components of 2D projectile motion are decoupled — linked only by the common time variable.**

Key idea: horizontal and vertical motions in 2D are independent. The vertical motion is identical for both balls (free fall from rest), giving identical fall times.

SHORT\_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-05

Q3. Take a ball and project it horizontally from the top of a building. Observe the projectile motion of the ball and explain with reasons which two quantities are constant throughout the projectile motion (when air resistance is negligible).

**Answer: Two quantities that are constant throughout projectile motion (air resistance neglected): (1) Horizontal velocity component ( $v_x$ ): there is no horizontal force acting on the projectile (gravity is purely vertical, air resistance is ignored). By Newton's first law, the horizontal velocity remains unchanged.  $v_x = u$  (the initial horizontal velocity). (2) Vertical acceleration component ( $a_y = -g$ ): the only force on the projectile is gravity, acting straight downward. So the vertical acceleration is constantly  $g$  downward, regardless of the projectile's horizontal motion. Variables that CHANGE: the vertical velocity ( $v_y$  starts at zero and increases linearly under  $g$ ); the speed (combination of constant  $v_x$  and growing  $v_y$ ); the direction of motion (angle below horizontal increases as the projectile falls).**

Two constants: horizontal velocity (no horizontal force) and vertical acceleration ( $= g$ , only force is gravity).

SHORT\_ANSWER · EASY · 12-NIOS-PHY-MIP-06

Q4. Suppose you want to launch a projectile to land at a certain target (for instance, a cricket ball beyond the boundary). What are the important factors you have to calculate before launching the projectile? Analyse projectile motion to determine all these factors.

**Answer: Key factors to control / calculate for hitting a target:** (1) **Launch SPEED ( $u$ )** — sets the total energy and the maximum reachable distance. Range scales as  $u^2$ . For longer hits, swing harder. (2) **Launch ANGLE ( $\theta$  above horizontal)** — sets the trade-off between range and height. On level ground, range  $R = u^2 \sin 2\theta / g$  is maximised at  $\theta = 45^\circ$ . For a 'six' over the boundary you want low trajectory + long range, so  $\theta \approx 35\text{--}45^\circ$ . (3) **Landing height relative to launch height ( $\Delta h$ )** — if the boundary is at the same height as the bat, the simple formulas apply. If higher (e.g. a fielder catching mid-air), you must solve the trajectory equation  $y = x \tan \theta - g x^2 / (2 u^2 \cos^2 \theta)$  for the appropriate  $y$ . (4) **Wind direction and speed** — adds a velocity to the ball relative to the ground. With a tailwind, increase the angle slightly; with a headwind, hit harder. (5) **Air resistance** — reduces real-world range, especially for fast-moving balls. The  $45^\circ$  optimum becomes  $\sim 30\text{--}40^\circ$  for real balls (lower for faster balls). **Main equations (no air resistance, level ground):** Time of flight  $T = 2 u \sin \theta / g$  Maximum height  $H = u^2 \sin^2 \theta / (2g)$  Range  $R = u^2 \sin 2\theta / g$  Given a target at horizontal distance  $R$ , the batsman needs  $u^2 \sin 2\theta / g = R$ . Solving gives the  $(u, \theta)$  combinations that just reach the target. For maximum reach, swing at  $u_{\text{max}}$  with  $\theta = 45^\circ$ . To clear a specific obstacle height, the trajectory equation pins down the angle uniquely (or shows it's impossible).

Discuss launch speed, launch angle, height difference, and the three formulas. Mention that the optimal angle is  $45^\circ$  on level ground but adjusts for air resistance, wind and height differences.

LONG\_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-06

Q5. Suppose you are riding a bicycle along a circular path of the park near your house. Measure the radius of the circular path. Count the time to complete one round and calculate the total length of the circular path.

**Answer: Sample values (substitute your measured ones):** Radius  $r = 20$  m. Time per lap  $T = 40$  s. Length of circular path (circumference)  $= 2 \pi r = 2 \times 3.14159 \times 20 \approx 125.66$  m. Linear speed  $= \text{circumference} / \text{period} = 125.66 / 40 \approx 3.14$  m/s  $\approx 11.3$  km/h. Angular speed  $\omega = 2\pi / T = 2\pi / 40 \approx 0.157$  rad/s. Centripetal acceleration  $a = \omega^2 r = 0.157^2 \times 20 \approx 0.493$  m/s<sup>2</sup> (or  $v^2/r = 9.87 / 20 \approx 0.49$  m/s<sup>2</sup>). For a 60-kg cyclist, the centripetal force required is  $F = m v^2 / r = 60 \times 9.87 / 20 \approx 30$  N — supplied by friction between the tyres and the ground (with the bike leaning inward to balance torques).

Apply circumference  $= 2\pi r$ ; linear speed  $= 2\pi r/T$ ; angular speed  $= 2\pi/T$ ;  $a_c = v^2/r$ .

SHORT\_ANSWER · EASY · 12-NIOS-PHY-MIP-08

Q6. Take three stones and throw them at different angles such that they reach the same maximum height. Observe the trajectories of all three projectiles and calculate for all three (a) time of flight, (b) horizontal velocity component, (c) downward acceleration component. Are these the same for all three stones? Explain.

**Answer: Condition: three projectiles reach the same maximum height  $H$ . So they have the same vertical initial velocity component:**  $H = u_y^2 / (2g) \Rightarrow u_y = \sqrt{2gH}$  — same for all three. For different launch angles  $\theta_1 < \theta_2 < \theta_3$ , the three stones must have different LAUNCH SPEEDS  $u_i = u_y / \sin \theta_i$ . So  $u_x = u_i \cos \theta_i = u_y / \tan \theta_i$  — three DIFFERENT horizontal velocity components. (a) Time of flight  $T = 2 u_y / g$  = same for all three (depends only on  $u_y$ , which we made equal). (b) Horizontal velocity components are DIFFERENT ( $= u_y / \tan \theta_i$ ). The shallowest-angle stone has the largest horizontal speed. (c) Vertical downward acceleration component is  $g$  — same for all three (gravity is universal). So TIME OF FLIGHT and VERTICAL ACCELERATION are the same for all three stones (because both depend only on the vertical motion, which is identical when max heights are equal). HORIZONTAL VELOCITY differs between stones (steeper angle  $\rightarrow$  smaller  $v_x \rightarrow$  smaller range). The three trajectories all reach the same peak, all spend the same time aloft, but cover different horizontal distances — like three balls launched from a single cannon at different elevations to clear the same wall.

Same max height  $\Rightarrow$  same  $u_y \Rightarrow$  same  $T$  and same  $g$ . But different launch angles mean different  $u$  and different  $u_x$ , so different ranges.

LONG\_ANSWER · HARD · 12-NIOS-PHY-MIP-07

Q7. Derive the equation of the path of projectile motion and the equation of the trajectory.

**Answer: Set up: take the launch point as origin,  $x$  horizontal in the direction of motion,  $y$  vertical upward. Launch speed  $u$  at angle  $\theta$  above horizontal. Initial velocity components:  $u_x = u \cos \theta$  (constant),  $u_y = u \sin \theta$ . Accelerations:  $a_x = 0$  (no horizontal force),  $a_y = -g$  (gravity downward). Apply kinematic equations component-wise (each component obeys the 1D equations since  $a$  is constant):  $x(t) = u \cos \theta \cdot t$  ... (i)  $y(t) = u \sin \theta \cdot t - (1/2) g t^2$  ... (ii) From (i):  $t = x / (u \cos \theta)$ . Substitute into (ii):  $y = u \sin \theta \cdot (x / (u \cos \theta)) - (1/2) g (x / (u \cos \theta))^2 = x \tan \theta - (g x^2) / (2 u^2 \cos^2 \theta)$  This is the EQUATION OF TRAJECTORY of the projectile. It has the form  $y = A x - B x^2$ , which is the equation of a PARABOLA opening downward, with  $A = \tan \theta$  and  $B = g / (2 u^2 \cos^2 \theta)$ . Maximum height:  $dy/dx = 0 \Rightarrow x_{\max} = (u^2 \sin \theta \cos \theta) / g = (u^2 \sin 2\theta) / (2g) = R/2$ . So the maximum height is at half the range, as expected by symmetry. The trajectory equation contains all the projectile-motion physics: range (set  $y = 0$  and solve for  $x \neq 0$ ), time of flight, maximum height, all follow from it.**

Eliminate  $t$  between  $x(t)$  and  $y(t)$  to get  $y$  as a function of  $x$ . The result is  $y = x \tan \theta - g x^2 / (2 u^2 \cos^2 \theta)$ , a parabola.

LONG\_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-06

Q8. Observe your surroundings and give examples of uniform circular motion from day to day life. Explain applications of uniform circular motion in day to day life.

**Answer: Examples of uniform circular motion (constant speed in a circle):** (1) Tip of a fan blade — rotates at fixed speed in a circle. (2) Hands of a clock — second hand traces a circle in 60 seconds. (3) Earth's orbit around the Sun (very nearly circular at constant speed). (4) Moon's orbit around Earth. (5) A satellite in a circular orbit. (6) A stone tied to a string, whirled at constant speed. (7) The wheel of a moving bicycle (a point on the rim). (8) Centrifuge / spin dryer — spins clothes around the drum. **Applications:** (1) Centrifuges separate substances of different densities (cream from milk, plasma from blood) — centripetal force varies with density. (2) Satellites and GPS rely on circular (or near-circular) orbits with precisely controlled centripetal force from gravity. (3) Banking of railway tracks and roads on curves — uses geometry to supply part of the required centripetal force. (4) Washing machines extract water using centrifugal effect during spin cycle. (5) Amusement park rides — rotating swings, Ferris wheel. (6) Cyclotrons accelerate charged particles in spiral paths. In each case, the centripetal force is supplied by some real physical agent — tension, gravity, friction, normal force — never by a 'centrifugal' force in the lab frame.

Look for at least three valid examples of UCM and an explanation of at least one application using the centripetal-force concept.

SHORT\_ANSWER · EASY · 12-NIOS-PHY-MIP-08

Q9. A truck goes around a circular track of radius  $R$  at speed  $v$  so that it makes one circuit every  $T$  seconds. It experiences a centripetal acceleration of magnitude  $p$ . If the truck now goes around a different circular track of radius  $4R$ , so that it takes time  $T/2$  to go around once, what is the magnitude of its centripetal acceleration in the second case?

**Answer: Centripetal acceleration in terms of period and radius:**  $a = \omega^2 r = (2\pi/T)^2 r = 4\pi^2 r / T^2$ . **Case 1:**  $p = 4\pi^2 R / T^2$ . **Case 2:**  $r' = 4R$ ,  $T' = T/2$ .  $a' = 4\pi^2 (4R) / (T/2)^2 = 4\pi^2 \times 4R \times 4 / T^2 = 16 \times (4\pi^2 R / T^2) = 16p$ . **The centripetal acceleration in the second case is 16 times the first case. Intuition:**  $a \propto r / T^2$ . **Radius increased by 4 (multiplies  $a$  by 4); period halved ( $T^2$  goes down by 4, so  $a$  goes UP by 4). Combined:  $4 \times 4 = 16$ .**

Use  $a = 4\pi^2 r / T^2$ . Each factor change is squared for  $T$  (denominator).

SHORT\_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-08

Q10. Explain in your own words what is meant by centripetal acceleration. Derive an expression for the same.

**Answer:** Centripetal acceleration is the acceleration of a particle moving in a circle at constant speed (uniform circular motion). It is directed at every instant toward the centre of the circle. The name 'centripetal' is Latin for 'centre-seeking'. Even though the SPEED of the particle is constant, the DIRECTION of its velocity is continuously changing — so the velocity vector is continuously changing, which is exactly what acceleration measures. Derivation: consider a particle moving on a circle of radius  $r$  at constant speed  $v$ . At time  $t$  the particle is at position A with velocity vector  $\mathbf{v}_A$  tangent to the circle; at time  $t + \Delta t$  it has moved to position B with velocity  $\mathbf{v}_B$  tangent at B. The position vector has rotated through a small angle  $\Delta\theta = v \Delta t / r$  (because the arc AB has length  $v \Delta t$  and the radius is  $r$ ). The two velocity vectors  $\mathbf{v}_A$  and  $\mathbf{v}_B$  have the same magnitude  $v$  but point in different directions. Drawing them from a common point gives a small isosceles triangle: the side connecting the tips is  $\Delta\mathbf{v}$ , with length  $|\Delta\mathbf{v}| \approx v \cdot \Delta\theta$  for small  $\Delta\theta$  (the chord of a small angle on a circle of radius  $v$ ). So  $|\Delta\mathbf{v}| = v \times (v \Delta t / r) = v^2 \Delta t / r$ . Magnitude of acceleration:  $a = |\Delta\mathbf{v}| / \Delta t = v^2 / r$ . Direction: as  $\Delta t \rightarrow 0$ ,  $\Delta\mathbf{v}$  becomes perpendicular to  $\mathbf{v}_A$  and points toward the centre of the circle. Therefore  $\mathbf{a}$  is directed toward the centre. Final result:  $\mathbf{a} = v^2 / r$  toward the centre. Equivalently, using  $v = \omega r$  where  $\omega$  is the angular speed:  $\mathbf{a} = \omega^2 r$ . This is the centripetal acceleration. The corresponding net inward force (centripetal force) is  $F = m a = m v^2 / r$ .

Conceptual: acceleration tracks the change in velocity VECTOR, not just speed. Geometrically derive  $|\Delta\mathbf{v}|/\Delta t = v^2/r$  in the limit of small  $\Delta t$ .

LONG\_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-08